PROPOSAL FOR A NEW SERVICE USING AI: ACCELERATING TELECOM BUSINESS WITH DATA ANALYSIS

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CURRENT STATE OF THE TELECOM INDUSTRY

As highlighted by Schwarz (2026), the telecommunications industry is currently evolving from providing legacy connectivity into becoming providers of integrated technology and managed customer experiences.

While the consumer market provides the bulk of revenue, Schwarz noted that consumer growth has been significantly eroded by low-cost providers and over-the-top (OTT) internet services.

To retain and reinvigorate consumer growth, telecom companies are focusing on transforming historically weak customer experiences.

By using predictive analytics, telecom companies can identify early churn signals, such as reduced usage or changing payment patterns, to reach out with service recovery options before a customer even complains (Schwarz, 2026).

THE GOAL: TARGETED PROACTIVE RETENTION

The Industry Challenge

With competition in the telecommunication industry, customers have a tremendous number of choices, making effective churn prediction models critical for business survival.

Our Objective

To move away from traditional methods and apply a novel retention technique called "targeted proactive retention," which combines Machine Learning and Big Data Analytics to address churn effectively (Mishachandar & Kumar, 2018).

The Strategy

By predicting customer churn behavior in advance, the company can fully comprehend the associated risks, specifically by forecasting the possibility and the time of occurrence of a churn event, to retain them (Mishachandar & Anil Kumar, 2018).

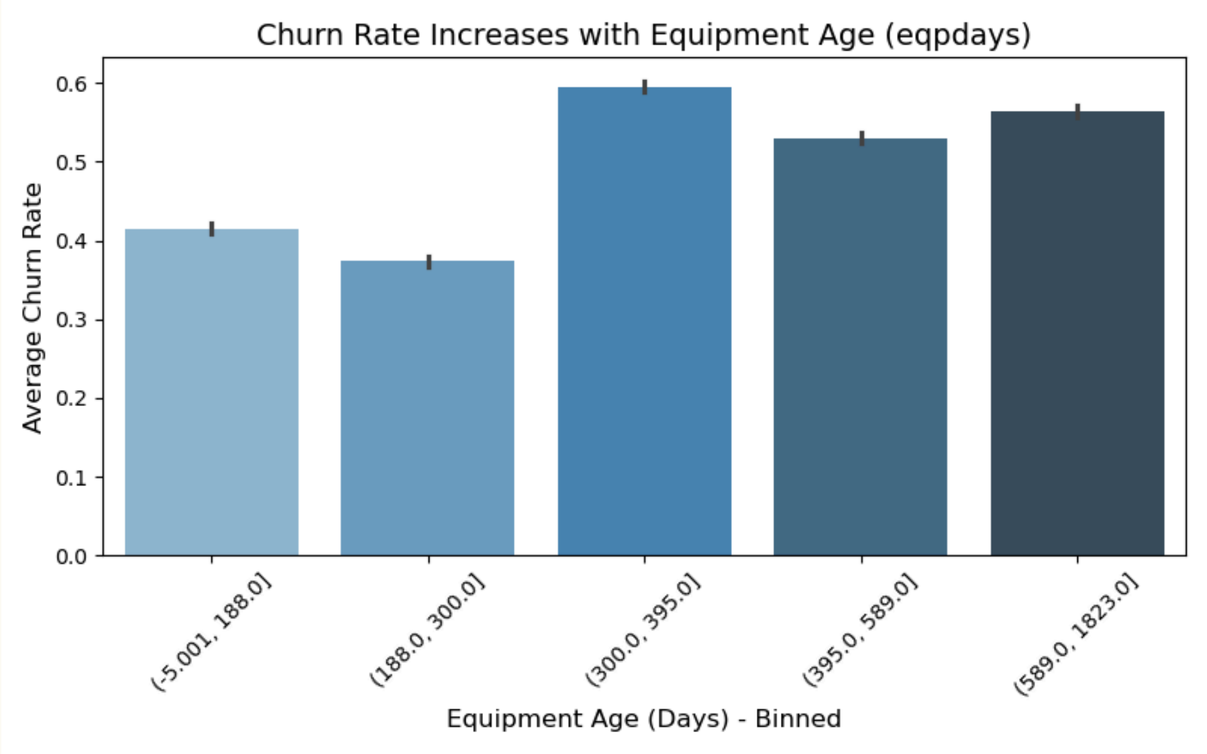
EXPLORATORY DATA ANALYSIS (EDA)

One-Year Itch Phenomenon

- Based on our historical data, the churn rate jumps from ~38% to nearly 60% once a customer's equipment age (eqpdays) reaches the 300 to 395-day mark.

- **Business Insight:** Customers are most vulnerable to leaving right around their 10th to 12th month. This likely corresponds to the end of initial 1-year contracts, the perceived "aging" of their handset, or competitors swooping in with new phone offers.

- **Actionable Strategy:** Instead of waiting for them to leave, we must launch Proactive Device Upgrade Campaigns specifically targeting users exactly at their 9th month of service, securing their loyalty before they hit the high-risk window.



MACHINE LEARNING MODEL

- **Method:** Random Forest Classifier (AI Ensemble Learning)
- **Target:** Churn (1 = Left the service, 0 = Stayed)
- **Evaluation Metric:** ROC-AUC Score
- **Model Score:** 0.6584



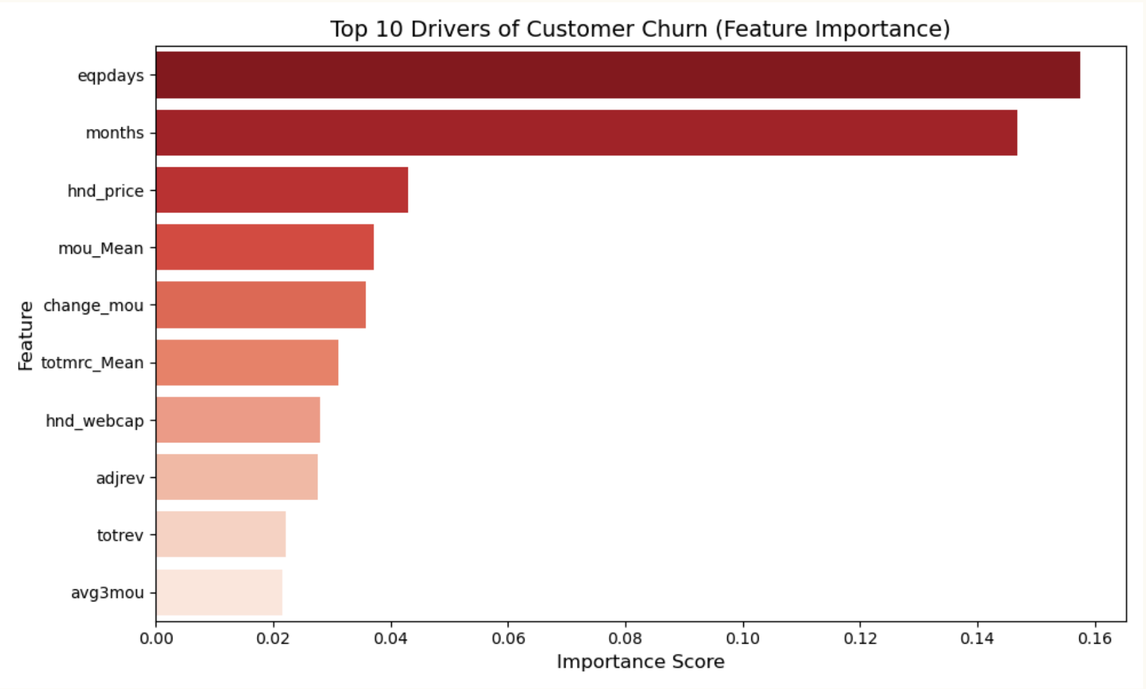
IMPORTANT FEATURES FOR PREDICTION

Hardware & Lifecycle Over Raw Usage

- Our Machine Learning model revealed that Equipment Age (eqpdays) and Tenure (months) are the absolute strongest predictors of churn7, outweighing even the customers' monthly charges or total revenue. Handset price and web capability also rank in the top 10.

- **Behavioral Shifts Matter:** The feature change_mou (percentage change in minutes used) is a warning sign. When customers begin calling less than their previous 3-month average, they are silently disengaging.

- **Business Insight:** The AI proves that customer retention is about hardware satisfaction and not just about the discounts given. By feeding these top 10 features into our Profit Maximization System (PMS), we can instantly flag customers who have aging devices and dropping usage, allowing us to intervene with pinpoint accuracy.



OUR BUSINESS PROPOSAL



AI-Driven Retention System

Risk Management

The AI automatically scores all active users monthly based on their usage behavior.

Estimation of High-Risk Customers

The system flags the top 10% of customers most likely to churn.

Targeted Action

Automatically send personalized offers (e.g., free phone upgrades or discounted data plans) to these specific users.

BUSINESS IMPACT & QUOTATION

Expected ROI

By accurately targeting high-risk customers and reducing the monthly churn rate by 5%, we can prevent the loss of millions of yen in recurring revenue and reduce the need for large-scale new acquisitions.

System Integration Cost

We propose a one-time fee of 30 Million Yen for the AI model delivery.







LONG-TERM SUPPORT

For 2.5 Million Yen / month, our team will provide time-series analysis and model updates adapting to economic trends.

REFERENCES

[1] Mishachandar, B., & Anil Kumar, K. (2018). Predicting customer churn using targeted proactive retention. International Journal of Engineering & Technology, 7(2.27), 69. https://doi.org/10.14419/ijet.v7i2.27.10180

[2] Schwarz, L. (2026, January 6). 12 Crucial Trends Driving the Future of the Telecom Industry. Oracle NetSuite. https://www.netsuite.com/portal/resource/articles/erp/future-of-telecom-industry.shtml